

Total marks-120**Attempt Questions 1-8****All questions are of equal value**

Answer each question in a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (15 Marks)

a) $\int \frac{xdx}{\sqrt{9-4x^2}}$ 2

b) $\int \frac{dx}{\sqrt{9-4x^2}}$ 2

c) Use integration by parts to evaluate $\int_1^e x^3 \ln x dx$ 3

d) (i) Find real numbers a, b and c such that $\frac{5x^2 - 4x - 9}{(x-2)(x^2-3)} = \frac{a}{x-2} + \frac{bx+c}{x^2-3}$ 2

(ii) Hence show that $\int_3^4 \frac{5x^2 - 4x - 9}{(x-2)(x^2-3)} = \ln \frac{52}{3}$ 2

e) $\int \sec^3 x \tan x dx$ 2

f) $\int \frac{dx}{x^2 + 4x + 13}$ 2

Question 2 (15 marks)

a) Let $z = 2 + i$ and $w = 3 - 4i$, find

(i) z^2 1

(ii) $\frac{1}{z}$ 1

(iii) $w\bar{z}$ 1

b) (i) Express $1 - \sqrt{3}i$ in mod arg form 2

(ii) Hence find $(1 - \sqrt{3}i)^5$ 1

(iii) Express $(1 - \sqrt{3}i)^5$ in the form $x + yi$ where x and y are real 2

c) If u and v are two non zero complex numbers. Show that if $\frac{u}{v} = ik$ for some $k \in \mathbb{R}$

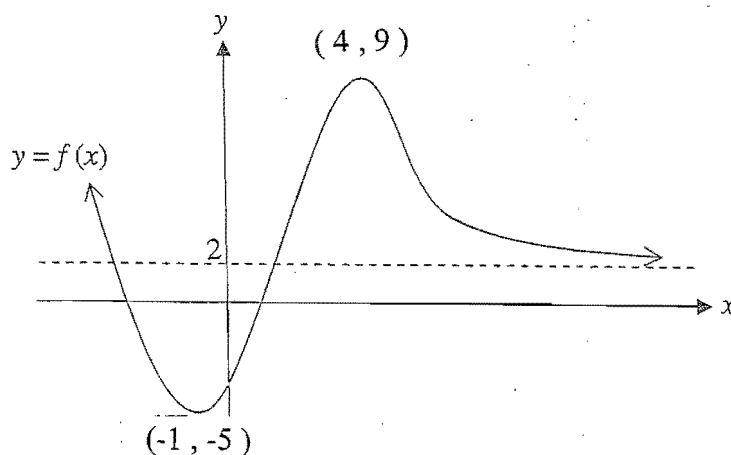
(i) $\bar{u}v + \bar{v}u = 0$ 2

(ii) If $\bar{u}v + \bar{v}u = 0$ what is the relationship between $\arg v$ and $\arg u$ 2

d) If ω is a complex root of the equation $z^3 = 1$

(i) Show that $1 + \omega + \omega^2 = 0$ 1

(ii) Find the value of $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$ 2

Question 3 (15 Marks)

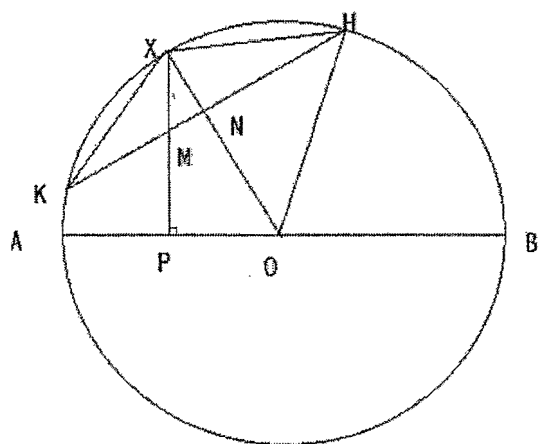
- a) The graph of $y = f(x)$ is shown above. It has been reproduced for you on pages 9 and 10, detach these pages and draw neat sketches of the following. Include these pages in your solutions. The point of intersection of $f(x)$ and the asymptote is $(1, 2)$.

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = f(|x|)$ 2

(iii) $y = f'(x)$ 2

(iv) $y = f\left(\frac{1}{x}\right)$ 2



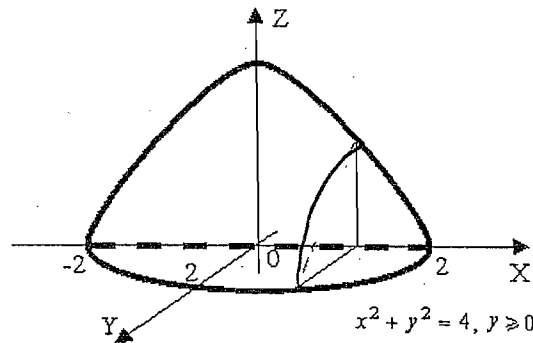
- b) The circle above has diameter AB and centre O. KH is a chord to the circle and X is a point on the circumference such that $KX = XH$. XP is the perpendicular from P to AB. Prove that PNMO is a cyclic quadrilateral. 3

c) (i) Find the square root of $-8 - 8\sqrt{3}i$ 2

(ii) Hence solve the quadratic equation $x^2 - 2\sqrt{2}ix + 2\sqrt{3}i = 0$ 2

Question 4 (15 Marks)

- a) The solid shown has a semicircular base of radius 2 units. Vertical cross sections perpendicular to the diameter of the circle are quarter circles.



- (i) By slicing at right angles to the x -axis show that the volume is given by

$$V = \frac{\pi}{2} \int_0^2 4 - x^2 dx \quad 2$$

- (ii) Find the volume 2

- b) The region bounded by the curve $y = \sin^{-1} x$ and the x -axis in the first quadrant is rotated about the line $y = -1$. Using the method of cylindrical shells find the volume of the shape formed. 4

- c) Let α, β and γ be the roots of the cubic equation $x^3 - 5x^2 + 13x - 7 = 0$.

- (i) Find the polynomial with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ 2

- (ii) Find the polynomial with roots α^2, β^2 and γ^2 2

- d) (i) Prove the identity $\sin(a+b)\theta + \sin(a-b)\theta = 2 \sin a\theta \cos b\theta$ 1

- (ii) Hence find $\int \sin 4\theta \cos 2\theta d\theta$ 2

Question 5 (15 Marks)

a) Given the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Find:

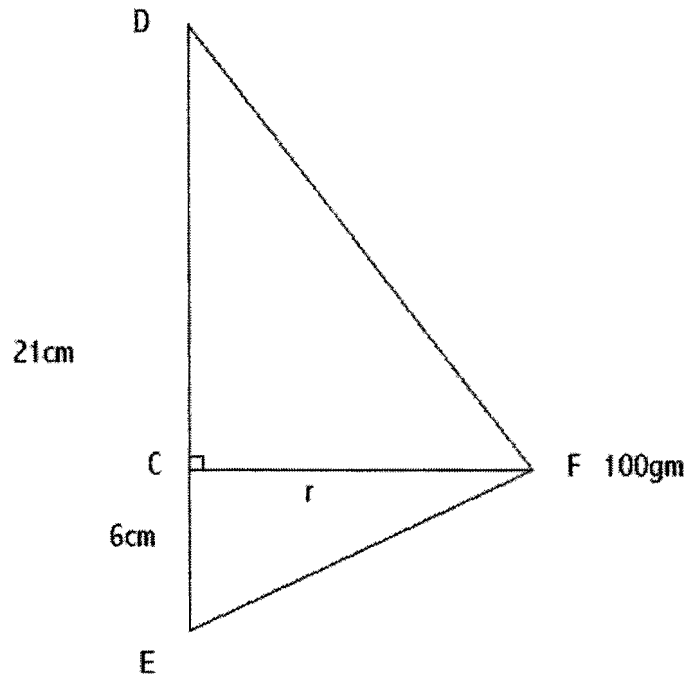
- (i) the eccentricity. 2
- (ii) The coordinates of the foci 1
- (iii) The equation of the directrices 1
- (iv) Sketch the ellipse showing all essential features. 2

b) Given the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$

- (i) Show that the point P with coordinates $(3 \sec \theta, 2 \tan \theta)$ lies on the hyperbola 1
- (ii) Find the equation of the normal to the hyperbola at P. 2
- (iii) Find the equation of the tangent to the hyperbola at P. 2
- (iv) The tangent at P cuts the asymptotes at L and M. Find the coordinates of L and M. 2
- (v) Show that P is the mid point of LM. 2

Question 6. (15 Marks)

a)



A light inelastic string of length 27cm is attached to two points D and E on the vertical shaft DE, distance 21cm apart, E being vertically below D. F is a smooth ring of mass 100gms threaded on the string. The system is such that F moves with constant speed in a horizontal circle 6cm above E.

- | | | |
|-------|--------------------------------------|---|
| (i) | Find the lengths of DF, FE and r . | 3 |
| (ii) | Find the tension in the string. | 2 |
| (iii) | Find the angular speed of F about DE | 2 |

b) A bullet is fired vertically into the air with a speed of 800m/s. In the air the bullet experiences

air resistance equal to $\frac{mv}{5}$ as well as gravity.

- | | | |
|-------|--|---|
| (i) | Find the height reached to the nearest metre. | 2 |
| (ii) | The time taken to achieve this height. | 2 |
| (iii) | As the bullet returns to the ground it is subject to the same forces,
Find the terminal velocity. | 2 |

c)	Solve for x $\tan^{-1} 3x - \tan^{-1} 2x = \tan^{-1} \frac{1}{5}$	2
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Question 7 (15 Marks)

a) The cubic equation $x^3 - 3x - 1 = 0$ is solved in two steps. Firstly let $x = u + v$ and secondly solve the quadratic equation $\lambda^2 - \lambda + 1 = 0$ the roots of which are u^3 and v^3 .

- (i) Solve the quadratic equation for u^3 and v^3 . 1
- (ii) Use De Moivre's theorem to find the cube roots with the arguments of least magnitude. 3
- (iii) Find the value of x leave in trigonometric form. 1

b) Let $I_n = \int_0^1 x^n \sqrt{1-x} dx$ $n = 0, 1, 2, 3 \dots$

- (i) Show that $I_n = \frac{2n}{2n+3} I_{n-1}$ 2
- (ii) Hence evaluate $\int_0^1 x^3 \sqrt{1-x} dx$ 2
- (iii) Show that $I_n = \frac{n!(n+1)!}{(2n+3)!} 4^{n+1}$ 2

c) The curves $y = \cos x$ and $y = \tan x$ intersect at a point P whose x coordinate is α

- (i) Show that the curves intersect at right angles at P. 2
- (ii) Show that $\sec^2 \alpha = \frac{1+\sqrt{5}}{2}$ 2

Question 8 (15 Marks)

- a) If $U_1 = \sqrt{2}$ and $U_n = \sqrt{2 + U_{n-1}}$ Prove by Mathematical Induction that

$$U_n < \sqrt{2} + 1 \text{ for all } n. \quad 3$$

- b) (i) Sketch the graph of $y = \frac{1}{x}$. With the aid of your sketch, show that for any

$$\text{positive number } u, \quad \frac{u}{1+u} < \int_1^{1+u} \frac{1}{x} dx < u \quad 1$$

- (ii) Deduce from (i) that $\frac{1}{1+r} < \ln \frac{r+1}{r} < \frac{1}{r}$, where $r > 0$ 1

- (iii) Let $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots + \frac{1}{n} - \ln n$. By using (ii) show that

$$\frac{1}{n} < a_n < 1 \quad 2$$

- c) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 2

- (ii) Hence find the value of $\int_0^{\pi} x \sin x \, dx$ 2

- d) (i) Show that the gradient function for $x^2 + y^2 + xy = 12$ is

$$\frac{dy}{dx} = \frac{-(2x+y)}{2y+x} \quad 2$$

- (ii) Find the coordinates of the stationary points of this function 1

- (iii) Find the coordinates of the points of contact of any vertical tangents. 1

(a) $I = \int \frac{x dx}{\sqrt{9-4x^2}}$

let $u = 9-4x^2$

$\frac{du}{dx} = -8x$

$du = -8x dx$

$I = -\frac{1}{8} \int \frac{-8x dx}{\sqrt{9-4x^2}}$

$= -\frac{1}{8} \int \frac{du}{\sqrt{u}}$

$= -\frac{1}{8} (2u^{1/2}) + C$

$= -\frac{1}{4} \sqrt{9-4x^2} + C$ (2)

(b) $I = \int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \int \frac{dv}{\sqrt{9/4 - v^2}}$
 $= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$ (2)

(c) $I = \int_1^e x^3 \ln x dx$
 $\int_1^e dv u = [uv]_1^e - \int_1^e v du$

let $u = \ln x$ $dv = x^3$
 $du = \frac{1}{x}$ $v = \frac{x^4}{4}$

$= \left[(\ln x) \frac{x^4}{4} \right]_1^e - \int_1^e \frac{x^4}{4} \frac{1}{x} dx$

$= \frac{e^4}{4} - \int_1^e \frac{x^3}{4} dx$

$= \frac{e^4}{4} - \left[\frac{x^4}{16} \right]_1^e$

$= \frac{e^4}{4} - \frac{e^4}{16} + \frac{1}{16}$

$= \frac{3e^4}{16} + \frac{1}{16} = \frac{1}{16} (3e^4 + 1)$ (3)

(d) (i) $\frac{5x^2 - 6x - 9}{(x-2)(x^2-3)} = \frac{a}{x-2} + \frac{bx+c}{x^2-3}$

$\therefore 5x^2 - 6x - 9 = a(x^2-3) + (bx+c)(x-2)$

let $x=2$ $a=3$

$x=0$ $c=0$ (2)

by coefficient of x^2 $b=2$

(ii) $\int_3^4 \frac{5x^2 - 6x - 9}{(x-2)(x^2-3)} dx = \int_3^4 \left(\frac{3}{x-2} + \frac{2x}{x^2-3} \right) dx$

$= \left[3 \ln(x-2) + \ln(x^2-3) \right]_3^4$

$= \left[3 \ln 2 + \ln(13) - 3 \ln 1 - \ln 6 \right]$

$= \ln \frac{8 \times 13}{6} = \ln \frac{4 \times 13}{3}$

$= \ln \frac{52}{3}$ (2)

(e) $\int \sec^2 x \tan x dx$

let $u = \sec x$

$\frac{du}{dx} = \sec x \tan x$

$\int u^2 du = \frac{u^3}{3} + C$

$= \frac{\sec^3 x}{3} + C$ (2)

(f) $\int \frac{dx}{x^2 \sqrt{4x+9}} = \int \frac{dx}{x^2 \sqrt{4x+9}}$

$= \int \frac{dx}{(x+2)^2 + 3^2}$

$= \frac{1}{3} \tan^{-1} \frac{x+2}{3} + C$ (2)

Q2.

(a) (i) $z^2 = (2+i)^2 = 3+4i$ (1)

(ii) $\frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{2-i}{(2+i)(2-i)} = \frac{2-i}{5}$ (1)

$= \frac{2-i}{5} = \frac{2}{5} - \frac{i}{5}$ (2)

(iii) $\omega \bar{z} = (3-4i)(2-i) = 2-11i$ (1)

(b) (i) $1-\sqrt{3}i \pmod{\sqrt{1^2+(-\sqrt{3})^2}}$
 $\pmod{2} = 2$ (2)

$\arg = \tan^{-1} \frac{-\sqrt{3}}{1} = -\pi/3$

(ii) $(1-\sqrt{3}i)^5 = 2^5 \cos 5(-\pi/3) = 32 \cos -5\pi/3 = 32 \cos \pi/3$ (1)

better

(iii) $(1-\sqrt{3}i)^5 = 32(\cos \pi/3 + i \sin \pi/3) = 16 + 16\sqrt{3}i$ (2)

(c) (i) $\bar{u}v + \bar{v}u = 0$

$\frac{u}{v} = ik$

$\therefore \left(\frac{u}{v}\right) = -ik$

$\frac{\bar{u}}{\bar{v}} = -ik$

$\frac{\bar{u}}{u} = \frac{-1}{ik}$

$\frac{u}{v} \times \frac{\bar{v}}{\bar{u}} = -1$

$\frac{u\bar{v}}{u\bar{v}} = -v\bar{u}$

$u\bar{v} + v\bar{u} = 0$

(ii) $\bar{u}v + \bar{v}u = 0$
 $\therefore \frac{u}{v} = ik$ (2)

$\arg \frac{u}{v} = ik$

$\arg \frac{u}{v} = \pm \pi/2$

$\arg u - \arg v = \pm \pi/2$
 $\therefore$ Difference of arg u and arg v is $\pi/2$ (2)

(d) (i) $z^3 = 1$

$z^3 - 1 = 0$

$(z-1)(z^2+z+1) = 0$
 $\therefore z=1$ is a root

$(\omega-1)(\omega^2+\omega+1) = 0$
 but ω is complex $\therefore \omega-1 \neq 0$

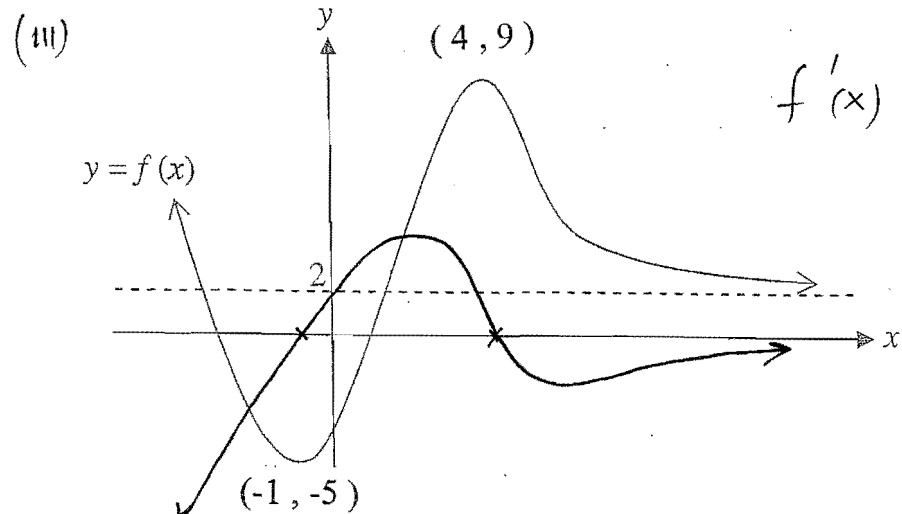
$\therefore \omega^2 + \omega + 1 = 0$
 or by sum of roots (1)

(ii) $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8) = (1+\omega)(1+\omega^2)(1+\omega)(1+\omega^2)$

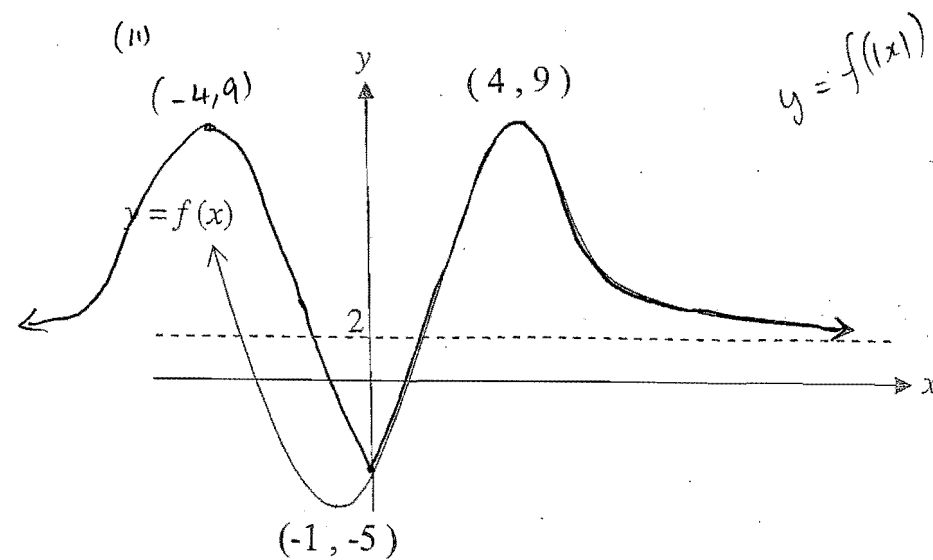
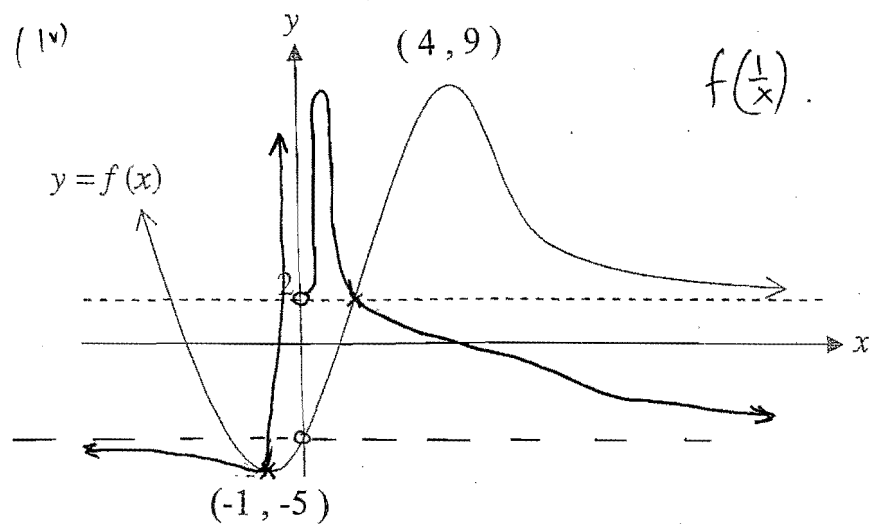
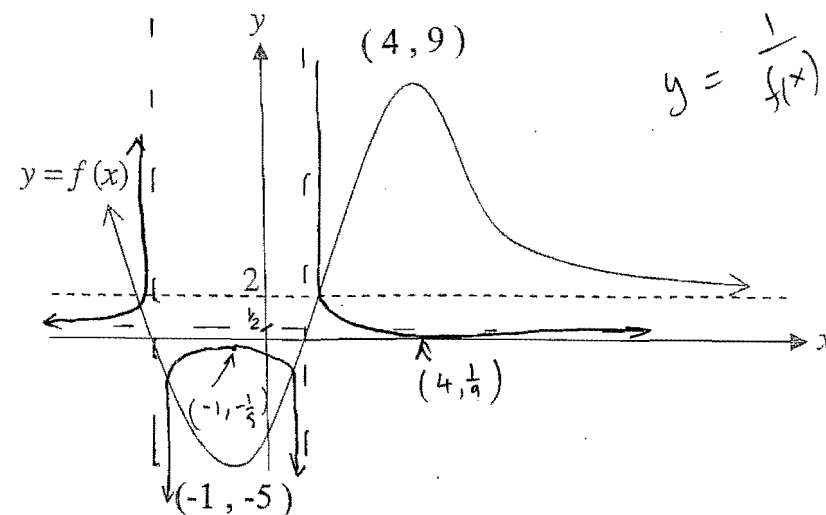
$= \omega^4 = \omega(\omega^3) = \omega$

$= \left\{ (1+\omega)(1+\omega^2) \right\}^2$

$= (1+\omega+\omega^2+\omega^3+\omega^2+\omega) = 1^2 = 1$ (2)



(iv)



Question 3.

(b) $\angle XOY = 2\angle XKH$ (2)
(angle at centre)

$\angle XHK = \angle XKH$ (ΔXKH is isosceles)

$OX = OH$ (radii)

$\therefore \angle OXH = \angle OHX$

$\angle XOY + 2\angle OXH = 180$

$\therefore 2\angle XKH + 2\angle OXH = 180$ (1)

$\angle XKH + \angle OXH = 90^\circ$

$\angle XHK + \angle OXH = 90^\circ$ (2)

$\therefore \angle XNH = 90^\circ$ ($\angle$ of ΔXNH)

$\therefore PNMO$ is cyclic (3)

$\angle NPO + \angle ONM = 180^\circ$

(c) (i) $\sqrt{-8-8\sqrt{3}i} = a+bi$

$-8-8\sqrt{3}i = a^2-b^2+2abi$

$-8 = a^2-b^2$

$-4\sqrt{3} = ab$ (2)

where $a = \pm 2$ $b = \pm 2\sqrt{3}$

(ii) $x^2 - 2\sqrt{2}ix + 2\sqrt{3}i = 0$

$x = \frac{2\sqrt{2}i \pm \sqrt{-8-8\sqrt{3}i}}{2}$

$x = \frac{2\sqrt{2}i \pm (2-2\sqrt{3}i)}{2}$

$x = \frac{2+2\sqrt{2}i-2\sqrt{3}i}{2}, \frac{-2+2\sqrt{2}i+2\sqrt{3}i}{2}$

$x = 1+(\sqrt{2}-\sqrt{3})i, -1+(\sqrt{2}+\sqrt{3})i$ (2)

Question 4

(a) (i) $A = \frac{1}{4}\pi r^2$

$= \frac{1}{4}\pi y^2$

$\delta V = \frac{1}{4}\pi y^2 \delta x$

$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^x \frac{1}{4}\pi y^2 \delta x$

$V = \frac{1}{2}\pi \int_0^2 y^2 dx$

but $x^2+y^2=4$

$y^2=4-x^2$

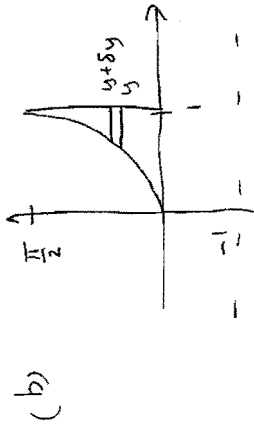
$V = \frac{\pi}{2} \int_0^2 4-x^2 dx$ (2)

(ii) $V = \frac{\pi}{2} \int_0^2 4-x^2 dx$

$= \frac{\pi}{2} \left[4x - \frac{x^3}{3} \right]_0^2$

$= \frac{\pi}{2} \left[(8 - \frac{8}{3}) - (0) \right]$

$= 8\pi/3$ (2)



$\delta V = \text{Vol outer cylinder}$

$= \text{Volume cylinder}$

$= \pi R^2 h - \pi r^2 h$

$= \pi \left((y+\delta y)^2 - (y)^2 \right) (1-x)$

$= \pi (1-x) \left\{ (y+\delta y)^2 - y^2 \right\}$

$\delta V = 2\pi(1-x)(y+\delta y)\delta y$

$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y_1} (1-x)(y+\delta y)\delta y$

$V = 2\pi \int_0^{y_1} (1-x)(y+\delta y) dy$

but $y = \sqrt{4-x^2}$

$\therefore x = \sqrt{4-y^2}$

$V = 2\pi \int_0^{y_1} (1-\sqrt{4-y^2})(y+\delta y) dy$

$= 2\pi \int_0^{y_1} y + \delta y dy - 2\pi \int_0^{y_1} y \sqrt{4-y^2} dy$

$V_A = 2\pi \left[\frac{y^2}{2} + y + \cos y \right]_0^{y_1}$

$= 2\pi \left[\left(\frac{y_1^2}{2} + y_1 + 0 \right) - (0+1) \right]$

$= 2\pi \left[\frac{y_1^2}{2} + y_1 - 1 \right]$

$V_B = -2\pi \int_0^{y_1} y \sqrt{4-y^2} dy$

by parts

$= -2\pi \left[-\frac{y}{2} \sqrt{4-y^2} - \int_0^{y_1} -\cos y dy \right]$

$= -2\pi \left\{ (0) + [\sin y]_0^{y_1} \right\}$

$= -2\pi$

$V_{\text{tot}} = V_A + V_B$ (4)

$= 2\pi \left[\frac{y_1^2}{2} + y_1 - 2 \right]$

(c) $f(x) = x^3 - 5x^2 + 13x - 7$

$f\left(\frac{1}{x}\right) = \frac{1}{x^3} - \frac{5}{x^2} + \frac{13}{x} - 7$

$x^3 f\left(\frac{1}{x}\right) = 1 - 5x + 13x^2 - 7x^3$

$f(x) = 7x^3 - 13x^2 + 5x - 1$ (2)

Question 4 (cont.)

(ii) roots α^2, β^2 and γ^2
 consider the function
 $f(\sqrt{x}) = (\sqrt{x})^3 - 5\sqrt{x} + 13\sqrt{x} - 7$
 $0 = 2\sqrt{x} - 5\sqrt{x} + 13\sqrt{x} - 7$
 $(5x+7)^2 = (x\sqrt{x} + 13\sqrt{x})^2$
 $25x^2 + 70x + 49 = x^3 + 26x^2 + 169x$
 $0 = x^3 + x^2 + 99x^2 - 49$
 $f(x) = x^3 + x^2 + 99x^2 - 49$

(d) (i) $\sin(a+b)\theta$
 $= \sin a\theta \cos b\theta + \cos a\theta \sin b\theta$
 $\sin(a-b)\theta$
 $= \sin a\theta \cos b\theta - \cos a\theta \sin b\theta$
 $\therefore \sin(a+b)\theta + \sin(a-b)\theta$
 $= 2 \sin a\theta \cos b\theta$ (1)

(ii) $\int \sin 4\theta \cos 2\theta d\theta$
 $a=4 \quad b=2$
 $= \frac{1}{2} \int \sin 6\theta + \sin 2\theta d\theta$
 $= \frac{1}{2} \left[-\frac{\cos 6\theta}{6} - \frac{\cos 2\theta}{2} \right] + C$
 $= -\frac{1}{12} [\cos 6\theta + 3 \cos 2\theta] + C$ (2)

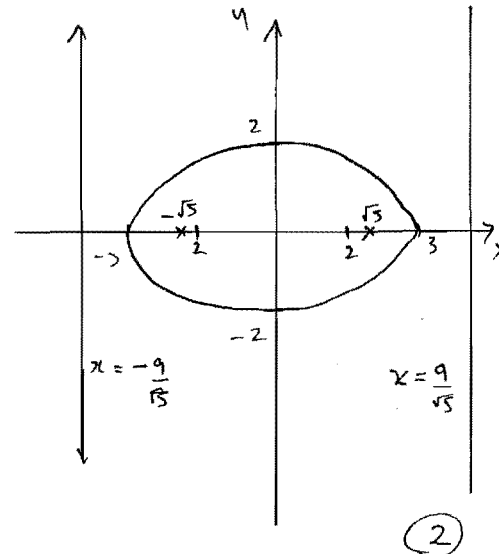
Question 5.

(a) $\frac{x^2}{9} + \frac{y^2}{4} = 1$

(i) $b^2 = a^2(1 - e^2)$
 $4 = 9(1 - e^2)$
 $\frac{4}{9} = 1 - e^2$
 $e = \sqrt{5}/3$ (2)

(ii) Foci $(\pm ae, 0)$
 $(\pm \sqrt{5}, 0)$ (1)

(iii) DIRECTRIX $x = \pm \frac{a}{e}$
 $x = \pm \frac{3}{\sqrt{5}/3}$
 $x = \pm \frac{9}{\sqrt{5}}$ (1)



(b) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

(i) $(\frac{3 \sec \theta}{9})^2 - (\frac{2 \tan \theta}{4})^2 = 1$
 $\sec^2 \theta - \tan^2 \theta = 1$
 $\sec^2 \theta = \tan^2 \theta + 1$ (true for all θ (pythag))

(ii) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

Diff $\frac{2x}{9} - \frac{2y}{4} \frac{dy}{dx} = 0$
 $\frac{dy}{dx} = \frac{4x}{9y}$
 $\frac{dy}{dx} = \frac{4(3 \sec \theta)}{9(2 \tan \theta)}$

$\therefore$ GRADIENT NORMAL $= -\frac{3 \tan \theta}{2 \sec \theta}$

Eqn of NORMAL

$y - 2 \tan \theta = -\frac{3 \tan \theta}{2 \sec \theta} (x - 3 \sec \theta)$

$\frac{y}{3 \tan \theta} - \frac{2}{3} = -\frac{x}{2 \sec \theta} + \frac{3}{2}$

$\frac{3x}{\sec \theta} + \frac{2y}{\tan \theta} = 13$ (2)

(iii) Eqn of TANGENT

$y - 2 \tan \theta = \frac{2 \sec \theta}{3 \tan \theta} (x - 3 \sec \theta)$

$3 \tan \theta y - 6 \tan^2 \theta = 2 \sec \theta x - 6 \sec \theta$

$6 \sec^2 \theta - 6 \tan^2 \theta = 2 \sec \theta x - 3 \tan \theta$

$6 = 2 \sec \theta x - 3 \tan \theta$

$1 = \frac{\sec \theta x}{3} - \frac{\tan \theta y}{2}$

Question 5 continued

(iv) Asymptotes

$$y = \pm \frac{b}{a} x$$

$$y = \frac{2}{3} x \quad y = -\frac{2}{3} x$$

$$1 = \frac{\sec \theta}{3} - \frac{\tan \theta}{2} \quad (2)$$

$$y = \frac{2}{3} x \quad (3)$$

$$L \left(\frac{3}{\sec \theta - \tan \theta}, \frac{2}{\sec \theta - \tan \theta} \right)$$

$$M \left(\frac{3}{\sec \theta + \tan \theta}, \frac{-2}{\sec \theta + \tan \theta} \right) \quad (2)$$

(v) Mid PT LM

$$x = \frac{1}{2} \left(\frac{3}{\sec \theta - \tan \theta} + \frac{3}{\sec \theta + \tan \theta} \right)$$

$$x = \frac{1}{2} \left(\frac{3 \sec \theta + 3 \tan \theta + 3 \sec \theta - 3 \tan \theta}{(\sec \theta - \tan \theta)(\sec \theta + \tan \theta)} \right)$$

$$x = \frac{3 \sec \theta}{\sec^2 \theta - \tan^2 \theta}$$

$$x = 3 \sec \theta$$

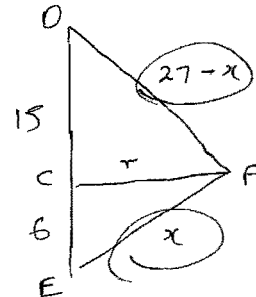
$$y = \frac{1}{2} \left(\frac{2}{\sec \theta - \tan \theta} + \frac{-2}{\sec \theta + \tan \theta} \right)$$

$$y = \frac{1}{2} \left(\frac{2 \sec \theta + 2 \tan \theta - 2 \sec \theta + 2 \tan \theta}{\sec^2 \theta - \tan^2 \theta} \right)$$

$$y = 2 \tan \theta$$

$\therefore$ P is mid pt of LM. (2)

Question 6.



$$DF^2 = r^2 + 15^2$$

$$FE^2 = r^2 + 6^2$$

$$(27-x)^2 = r^2 + 225 \quad (2)$$

$$x^2 = r^2 + 36 \quad (3)$$

$$729 - 54x + x^2 = r^2 + 225 \quad (4)$$

Subst (3)

$$729 - 54x + x^2 + 36 = r^2 + 225$$

$$54x = 540$$

$$x = 10$$

$$\therefore EF = 10 \text{ cm}$$

$$DF = 17 \text{ cm}$$

$$r = 8 \text{ cm} \quad (3)$$

Resolving vertically at F

$$Mg = T \cos \angle D - T \cos \angle E$$

$$0.1 \times 9.8 = T \frac{15}{17} - T \frac{3}{5}$$

$$T = 3.54 \text{ N.}$$

(2)

Resolving horizontally at F

$$m\omega^2 r = T \sin \angle D + T \sin \angle E$$

$$0.1 \times \omega^2 \times 0.08 = 3.54 \left(\frac{4}{5} + \frac{8}{17} \right)$$

$$\omega^2 = 563.5 \text{ rad/sec}$$

$$\omega = 23.7 \text{ rad/sec}$$

(b) (i) $F = ma$

$$ma = -\frac{mv}{s} - mg$$

$$a = -\left(\frac{v+sg}{s} \right)$$

FOR HEIGHT USE $a = v \frac{dv}{dx}$

$$v \frac{dv}{dx} = -\frac{(v+sg)}{s}$$

$$\frac{dv}{dx} = -\frac{v+sg}{sv}$$

$$\frac{dx}{dv} = -s \left(\frac{v}{v+sg} \right)$$

$$\int_0^H dx = -s \int_{800}^0 \frac{v}{v+sg} dv$$

$$H = s \int_0^{800} \left(1 - \frac{sg}{v+sg} \right) dv$$

$$H = s \left[v - sg \ln(v+sg) \right]_0^{800}$$

using $g = 10$.

$$H = s \left[(800 - 50 \ln 850) + 50 \ln 50 \right]$$

$$H = 4000 + 250 \ln \frac{50}{850}$$

$$H = \frac{3292 \text{ m}}{\quad} \quad (2)$$

(ii) FOR TIME USE $\frac{dv}{dt} = a$

$$\frac{dv}{dt} = -\frac{(v+sg)}{s}$$

$$\frac{dt}{dv} = -\frac{s}{v+sg}$$

$$dt = -\frac{s dv}{v+sg}$$

Question 6 cont.

$$\int_0^t dt = -5 \int_{800}^v \frac{dv}{v+5g}$$

$$t = 5 \left[\ln(v+5g) \right]_0^{800}$$

$$t = 5 (\ln 850 - \ln 50) \quad (2)$$

$$t = 5 \ln 17 \approx 14.17 \text{ secs}$$

(iii) $F = ma$

$$ma = Mg - \frac{mv}{5}$$

$$a = \frac{5g-v}{5}$$

FOR TERMINAL VELOCITY

$$a = 0$$

$$5g = v$$

$$v = 50 \text{ m/sec} \quad (2)$$

(c) $\tan^{-1} 3x - \tan^{-1} 2x = \tan^{-1} \frac{1}{5}$

$$\text{let } \alpha = \tan^{-1} 3x \quad 3x = \tan \alpha$$

$$\text{let } \beta = \tan^{-1} 2x \quad 2x = \tan \beta$$

$$\tan(\alpha - \beta) = \tan(\tan^{-1} 3x - \tan^{-1} 2x)$$

$$= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\tan(\tan^{-1} \frac{1}{5}) = \frac{3x - 2x}{1 + (3x)(2x)}$$

$$\frac{1}{5} = \frac{x}{1+6x^2}$$

$$1+6x^2+1 = 5x$$

$$0 = 6x^2 - 5x + 1$$

$$0 = (3x-1)(2x-1)$$

$$x = \frac{1}{2}, \frac{1}{3} \quad (2)$$

Question 7

(a) (i) $\lambda^2 - \lambda + 1 = 0$

$$\lambda = \frac{1 \pm \sqrt{1-4}}{2}$$

$$\lambda = \frac{1}{2} + \frac{\sqrt{3}i}{2}, \frac{1}{2} - \frac{\sqrt{3}i}{2} \quad (1)$$

(ii) $\lambda = 1 \cos \pi/3, 1 \cos -\pi/3$

$$\therefore u^3 = 1 \cos \pi/3, v^3 = 1 \cos -\pi/3$$

$$u = \cos \frac{2\pi k + \pi/3}{3}, v = \cos \frac{2\pi k - \pi/3}{3}$$

least arg

$$u = \cos \pi/9, v = \cos -\pi/9 \quad (3)$$

(iii) $x = u + v$

$$x = \cos \pi/9 + i \sin \pi/9 + \cos -\pi/9 + i \sin -\pi/9$$

$$x = 2 \cos \pi/9 \quad (1)$$

(b) $I_n = \int_0^1 x^n \sqrt{1-x} dx$

$$\text{let } u = x^n \quad \text{let } dv = (1-x)^{1/2}$$

$$\frac{du}{dx} = nx^{n-1} \quad v = -\frac{2}{3}(1-x)^{3/2}$$

$$\int_0^1 x^n \sqrt{1-x} dx = \left[uv \right]_0^1 - \int_0^1 v \frac{du}{dx}$$

$$= \left[x^n (1-x)^{3/2} \left(-\frac{2}{3} \right) \right]_0^1 + \frac{2}{3} \int_0^1 (1-x)^{3/2} (x^{n-1}) n$$

$$= 0 + \frac{2n}{3} \int_0^1 \sqrt{1-x} (1-x) x^{n-1}$$

$$= \frac{2n}{3} I_{n-1} - \frac{2n}{3} \int_0^1 \sqrt{1-x} x^n$$

$$\therefore \frac{2n+3}{3} I_n = \frac{2n}{3} I_{n-1}$$

$$\therefore I_n = \frac{2n}{2n+3} I_{n-1} \quad (2)$$

(ii) $\int_0^1 x^3 \sqrt{1-x} dx = \frac{6}{9} I_2$

$$I_2 = \frac{4}{7} I_1$$

$$I_1 = \frac{2}{5} I_0$$

$$I_0 = \int_0^1 \sqrt{1-x} dx = \left[-\frac{2}{3} (1-x)^{3/2} \right]_0^1 = \frac{2}{3}$$

$$I_3 = \frac{6}{9} \times \frac{4}{7} \times \frac{2}{5} \times \frac{2}{3} = \frac{32}{315} = \frac{4^4 3! 4!}{9!} \quad (2)$$

(iii) $I_n = \frac{2n}{2n+3} \times \frac{2n-2}{2n+1} \dots \frac{6}{9} \times \frac{4}{7} \times \frac{2}{5} \times \frac{2}{3}$

$$= \frac{2^{n+1} n!}{(2n+3)(2n+1) \dots 9 \times 7 \times 5 \times 3 \times 1}$$

$$= \frac{2^{n+1} n! \times (2n+2)(2n) \dots 10 \times 8 \times 6}{(2n+3)!}$$

$$= \frac{2^{n+1} n! (n+1)! 2^{n+1}}{(2n+3)!}$$

$$= \frac{n! (n+1)! 4^{n+1}}{(2n+3)!} \quad (2)$$

c) (ii) $\cos x = \tan x$

$$\cos x = \frac{\sin x}{\cos x}$$

$$\cos^2 x = \sin x$$

$$1 - \sin^2 x = \sin x$$

$$0 = \sin^2 x + \sin x - 1$$

Question 7 cont.

$$\sin x = -1 \pm \frac{\sqrt{1+u}}{2}$$

$$= -1 \pm \frac{\sqrt{5}}{2}$$

$$\sin x = -1 + \frac{\sqrt{5}}{2} \text{ acute } x$$

$$\text{but } \sin x = \cos^2 x$$

$$\cos^2 x = -1 + \frac{\sqrt{5}}{2}$$

$$\therefore \sec^2 x = \frac{2}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$$

$$= 2 \frac{(\sqrt{5}+1)}{4}$$

$$= \frac{1+\sqrt{5}}{2} \quad (2)$$

$$(i) \quad y' = -\sin x \quad y' = \sec^2 x$$

$$\text{when } x = \alpha$$

$$(-\sin \alpha)(\sec^2 \alpha) = \frac{1-\sqrt{5}}{2} \times \frac{1+\sqrt{5}}{2}$$

$$= \frac{1-5}{4} = -1$$

$\therefore$ product of gradient = -1

$\therefore$ $\cos x$ and $\tan x$

perpendicular at α (2)

Question 8.

$$a) \quad u_1 = \sqrt{2} \quad u_2 = \sqrt{2+u_{n-1}}$$

Step 1 Prove true for $n=1$

$$LHS = \sqrt{2}$$

$$RHS = \sqrt{2+1}$$

LHS < RHS true for $n=1$

Step 2. Assume true for $n=k$

$$\therefore u_k = \sqrt{2+u_{k-1}}$$

$$u_k < \sqrt{2+1}$$

Prove true for u_{k+1}

$$u_{k+1} = \sqrt{2+u_k}$$

$$< \sqrt{2+\sqrt{2}+1}$$

$$< \sqrt{3+2\sqrt{2}}$$

$$= \sqrt{(\sqrt{2}+1)^2}$$

$$= \sqrt{2}+1$$

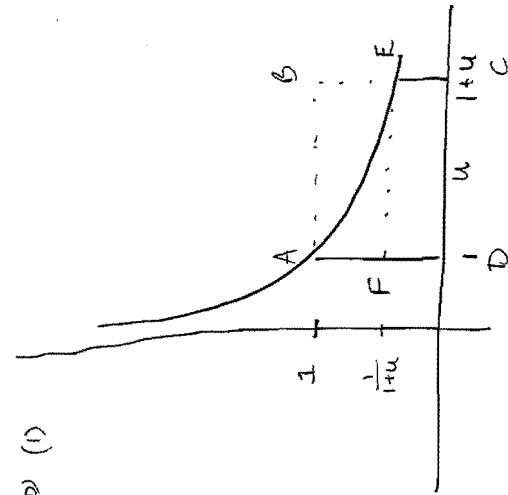
$$\therefore u_{k+1} < \sqrt{2}+1$$

Step 3 By the principle of

Mathematical Induction

true for all n (3)

(b) (i)



By inspection

$$A_{CDRE} < \int_{1+u}^{1+u} \frac{1}{x} < A_{ABCD}$$

$$u \times \frac{1}{1+u} < \int_{1+u}^{1+u} \frac{1}{x} < 1 \times u$$

$$\frac{u}{1+u} < \int_{1+u}^{1+u} \frac{1}{x} dx < u \quad (1)$$

(ii) Now since

$$\int_{1+u}^{1+u} \frac{1}{x} dx = [\ln x]_{1+u}^{1+u}$$

$$= \ln(1+u) - \ln 1$$

$$= \ln(1+u)$$

$$\frac{u}{1+u} < \ln(1+u) < u$$

$$\text{let } u = \frac{1}{r}$$

$$\frac{\frac{1}{r}}{1+\frac{1}{r}} < \ln(1+\frac{1}{r}) < \frac{1}{r}$$

$$\frac{1}{r+1} < \ln \frac{r+1}{r} < \frac{1}{r} \quad (1)$$

(iii) let $r = 1, 2, 3, \dots, n-1$

$$\frac{1}{2} < \ln 2 - \ln 1 < 1$$

$$\frac{1}{3} < \ln 3 - \ln 2 < \frac{1}{2}$$

$$\frac{1}{n} < \ln n - \ln(n-1) < \frac{1}{n-1}$$

adding all three sides

$$\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} < \ln n - \ln 1 < 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1}$$

so that

$$\frac{1}{n} < 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \ln n < 1$$

$$\frac{1}{n} < a_0 < 1 \quad (2)$$

$$(c) (i) \int_a^a f(x) dx$$

$$\text{let } x = a - u \quad \text{when } x = a$$

$$u = 0$$

$$\frac{dx}{du} = -1$$

$$\text{when } x = 0$$

$$u = a$$

Question 8 Cont.

$$\begin{aligned}\int_a^b f(x) dx &= \int_a^b f(a-u) - du \\ &= - \int_a^b f(a-u) du \\ &= \int_a^b f(a-u) du \\ \text{By change of variable} \\ &= \int_a^b f(a-x) dx\end{aligned}$$

$$\therefore \int_a^b f(x) dx = \int_a^b f(a-x) dx \quad (2)$$

$$\begin{aligned}\text{(ii)} \quad \therefore \int_0^\pi x \sin x \, dx \\ = \int_0^\pi (\pi-x) \sin(\pi-x) \, dx\end{aligned}$$

$$\text{N.B. } \sin(\pi-x) = \sin x$$

$$= \int_0^\pi \pi \sin x - x \sin x \, dx$$

$$\begin{aligned}\therefore 2 \int_0^\pi x \sin x \, dx &= \pi \int_0^\pi \sin x \\ \int_0^\pi x \sin x \, dx &= \frac{\pi}{2} [-\cos x]_0^\pi\end{aligned}$$

$$\begin{aligned}&= \frac{\pi}{2} [1+1] \\ &= \pi \cdot 1^2 \quad (2)\end{aligned}$$

$$\text{d)} \quad x^2 + y^2 + xy = 12$$

differentiate implicitly

$$2x + 2y \frac{dy}{dx} + x \frac{dy}{dx} + y = 0$$

$$(2y+x) \frac{dy}{dx} = -(2x+y)$$

$$\frac{dy}{dx} = - \frac{(2x+y)}{2y+x} \quad (2)$$

(ii) for stationary points

$$\begin{aligned}\frac{dy}{dx} &= 0 \\ 2x+y &= 0\end{aligned}$$

$$y = -2x$$

Subst into function

$$x^2 + (-2x)^2 + x(-2x) = 12$$

$$x^2 + 4x^2 - 2x^2 = 12$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = 2 \quad y = -4 \quad (2, -4)$$

$$x = -2 \quad y = 4 \quad (-2, 4) \quad (1)$$

(iii) for vertical tangent

$$2y+x=0$$

$$y = -x/2$$

$$x^2 + \left(-\frac{x}{2}\right)^2 + x\left(-\frac{x}{2}\right) = 12$$

$$x^2 + \frac{x^2}{4} - \frac{x^2}{2} = 12$$

$$\frac{3x^2}{4} = 12$$

$$x = \pm 4$$

$$(4, -2) \quad (-4, 2) \quad (1)$$